5(a) The direction of the induced emf is such as to produce an effect that opposes the change causing it [1]

(b)(i) X = 0.85 A [1]
Y = 2/ 0.040 [1]
= 160 rad s^{-1} [1]

(b)(ii) two cycles of a sinusoidal curve with a period of 0.040 s [1]
correct phase (i.e. V2 max / min at t = 0, 0.02, 0.04, 0.06 and 0.08 s, and V2 zero at t = 0.01, 0.03, 0.05, 0.07 s) [1]

maximum / minimum V_2 shown (consistently) at ± 6.5 V [1]

(b)(iii) (magnitude of) V_2 is proportional to rate of change of (magnetic) flux [1]

- V_2 is proportional to gradient of $I1$–t curve
- V_2 has maximum magnitude when $I1$–t curve is steepest
- V_2 is zero when $I1$–t curve is horizontal / a maximum or minimum
- V_2 changes sign when sign of gradient of $I1$–t curve changes

Any 2, give [1]

6(a)(i) From Fig 6.1 it can be observed that at specific wavelengths there were dips in the